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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device includes a semiconductor substrate, an interlayer insulating film that is provided on the semiconductor substrate, a first electrode that is provided on the interlayer insulating film, a second electrode that is provided below the semiconductor substrate, a plurality of gate regions extending from the interlayer insulating film in a first direction, which is a thickness direction of the semiconductor substrate, to reach the semiconductor substrate, provided in a second direction intersecting the first direction, and extending in a third direction intersecting the first direction and the second direction, and a plurality of plugs located between the gate regions in the second direction, longer in the second direction than in the third direction, spaced apart from each other in the third direction, and electrically connecting the first electrode to the semiconductor substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-032211, filed Mar. 4, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a semiconductor device.

BACKGROUND

[0003] A structure is known in which a plug electrically connecting an electrode and a semiconductor substrate opposed to each other via an insulating film is formed. The plug is formed by embedding, for example, W (tungsten) in an opening that is excavated by etching or the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1A is a plan view showing a semiconductor device according to a first embodiment.

[0005] FIG. 1B is an enlarged plan view showing a region Ra of the semiconductor device shown in FIG. 1A.

[0006] FIG. 2A is a cross-sectional view showing an A-A' cross section of the semiconductor device shown in FIG. 1B.

[0007] FIG. 2B is an enlarged cross-sectional view showing a B-B' cross section of the semiconductor device shown in FIG. 1B.

[0008] FIG. 2C is an enlarged cross-sectional view showing a C-C' cross section of the semiconductor device shown in FIG. 1B.

[0009] FIG. 2D is an enlarged cross-sectional view showing the C-C' cross section of another configuration of the semiconductor device shown in FIG. 1B.

[0010] FIG. 3 is an enlarged plan view showing a first modification of the semiconductor device according to the first embodiment.

[0011] FIG. 4 is an enlarged plan view showing a semiconductor device according to a first reference example.

[0012] FIG. 5 is a cross-sectional view showing a D-D' cross section of the semiconductor device shown in FIG. 4.

[0013] FIG. 6 is an enlarged plan view showing a semiconductor device according to a second reference example.

[0014] FIG. 7 is an enlarged plan view showing a semiconductor device according to a second embodiment.

[0015] FIG. 8A is a cross-sectional view showing an E-E' cross section of the semiconductor device shown in FIG. 7.

[0016] FIG. 8B is a cross-sectional view showing an F-F' cross section of the semiconductor device shown in FIG. 7.

[0017] FIG. 9 is an enlarged plan view showing a semiconductor device according to a third embodiment.

[0018] FIG. 10 is a cross-sectional view showing a G-G' cross section of the semiconductor device shown in FIG. 9.

[0019] FIGS. 11-16 are each a cross-sectional view corresponding to the A-A' cross section of the semiconductor device shown in FIG. 1, and illustrate a manufacturing method for the semiconductor device according to the first embodiment.

DETAILED DESCRIPTION

[0020] Embodiments provide a semiconductor device capable of optimizing current control therein and improving electrical characteristics thereof.

[0021] In general, according to one embodiment, a semiconductor device includes a semiconductor substrate, an interlayer insulating film that is provided on the semiconductor substrate, a first electrode that is provided on the interlayer insulating film, a second electrode that is provided below the semiconductor substrate, a plurality of gate regions extending from the interlayer insulating film in a first direction, which is a thickness direction of the semiconductor substrate, to reach the semiconductor substrate, provided in a second direction intersecting the first direction, and extending in a third direction intersecting the first direction and the second direction, and a plurality of plugs located between the gate regions in the second direction, longer in the second direction than in the third direction, spaced apart from each other in the third direction, and electrically connecting the first electrode to the semiconductor substrate.

[0022] Embodiments of the present disclosure will be described below with reference to the drawings.

[0023] The drawings are schematic or conceptual, and a relationship between the thickness and width of each part, a ratio between the sizes of parts, and the like are not necessarily the same as those of the actual ones. Even when the same part is shown, the dimensions and ratios may be different depending on the drawing.

[0024] In the present specification and the drawings, elements that are already described with respect to the previous drawings are given the same reference numerals, and detailed descriptions are omitted as appropriate.

[0025] A direction from a first semiconductor region 21 to a second semiconductor region 22 is referred to herein as a Z direction. The Z direction is the thickness direction of a semiconductor substrate 20. In addition, a direction intersecting the Z direction is referred to herein as an X direction, and a direction intersecting the X and Z directions is referred to herein as a Y direction. A semiconductor device 1 shown in FIG. 2 shows a cross-sectional view in an X-Z plane. Although the X, Y, and Z directions are shown in an orthogonal relationship in this embodiment, the directions are not limited to being orthogonal and may be in a mutually intersecting relationship.

[0026] In this specification, the positive direction of the Z direction is referred to as "upward", and the other side is referred to as "downward". The "upward" and "downward" directions are not limited to the direction of gravity or the direction when the semiconductor device is mounted.

[0027] In this specification, technical matters may be described using orthogonal coordinate axes of an X-axis, a Y-axis, and a Z-axis. The orthogonal coordinate axes merely specify relative positions of components, and are not intended to limit the directions recited in the claims to particular directions. For example, the Z-axis is not limited to a height direction with respect to the ground. The +Z-axis direction and the -Z-axis direction are opposite directions. When the Z-axis direction is written without indicating a positive or a negative, it means a direction parallel to the +Z-axis and the -Z-axis.

[0028] Further, in the following description, the notations n.sup.+, n, n.sup.-, p.sup.+, p, and p.sup.- represent the relative levels of impurity concentrations of respective conductivity types. That is, n.sup.+ indicates that the impurity concentration of an n-type is relatively higher than that of n, and n.sup.- indicates that the impurity concentration of an n-type is relatively lower than that of n. In addition, p.sup.+ indicates that the impurity concentration of a p-type is relatively higher than that of p, and p.sup.- indicates that the impurity concentration of a p-type is relatively lower than that

of p. An n.sup.+ -type and an n.sup.- -type may be simply written as an n-type, and a p.sup.+ -type and a p.sup.- -type may be simply written as a p-type.

First Embodiment

[0029] FIG. 1A is a schematic plan view showing an internal wiring structure of the semiconductor device **1** according to this embodiment. FIG. 1B is an enlarged plan view showing a region Ra shown in FIG. 1A. In FIGS. 1A and 1B, a first electrode **11** and an interlayer insulating film **50** are omitted in order to show the internal wiring structure.

[0030] FIG. 1A is a schematic top view of the semiconductor device **1** and an enlarged plan view of a portion of the top view. In an active region AR of the semiconductor device **1**, for example, an insulated gate bipolar transistor (IGBT) is formed. The active region AR is a region through which a current mainly flows when the semiconductor device **1** is in an ON state.

[0031] In the plan view of FIG. 1A, gate regions **30** are repeatedly provided in the X direction on the semiconductor substrate **20** and extend in the Y direction. A plurality of plugs **40** are provided in the semiconductor substrate **20** between the gate regions **30** provided in a plurality of rows in the X direction so as to be spaced apart from each other in the Y direction. Here, it is stated that the plugs **40** are provided between the gate regions **30**, but, for example, in a case where the plugs **40** are located on the outermost periphery outside the active region AR of the semiconductor device **1**, all of the plugs **40** do not need to be provided between the gate regions **30**.

[0032] The plan view of FIG. 1A schematically shows a portion of the active region AR, and a peripheral portion of the active region AR is not limited to the structure shown in FIG. 1A. In addition, for example, if an IGBT is formed in the active region AR, a portion where an emitter electrode is formed and a portion where a gate electrode is formed may have different structures.

[0033] FIG. 1B is an enlarged plan view of the region Ra shown in FIG. 1A. FIG. 1B shows the internal wiring structure. The gate region **30** includes, for example, a gate insulating film **31** and a gate electrode **32**. The gate electrode **32** is covered with the gate insulating film **31**. The gate electrodes **32** do not all need to be at the same potential. A dummy electrode may be included in a portion of the gate electrode **32** of the gate region **30**. In addition, the gate region **30** does not necessarily need to include the gate electrode **32**. A portion of the gate region **30** may be made of, for example, only an insulating material.

[0034] The semiconductor substrate **20** includes a first semiconductor region **21** of a first conductivity type (not shown in FIG. 1B), a second semiconductor region **22** of a second conductivity type, and a third semiconductor region **23** of a first conductivity type. The second semiconductor region **22** is, for example, a p-type base region. The third semiconductor region **23** is, for example, an n-type emitter region. The third semiconductor region **23** is selectively provided on the second semiconductor region **22**.

[0035] As shown in FIG. 1B, a plurality of third semiconductor regions **23** are provided in the Y direction, for example, between the gate regions **30**. For example, the length of the third semiconductor region **23** in the X direction is longer than the length thereof in the Y direction. The second semiconductor region **22** is located between the separated third semiconductor regions **23**. By providing the plurality of third semiconductor regions **23** in the Y direction, it is possible to control a contact area between the third semiconductor region **23** and the gate region **30** and disperse a flow of a current. As will be described below, when the semiconductor device **1** is in an ON state, a current flows through the third semiconductor region **23** in the vicinity of the gate region **30**. The flow of the current in an ON state can be controlled by the arrangement of the plurality of third semiconductor regions **23** formed in the Y direction. The third semiconductor regions **23** may be regions extending in the Y direction and formed continuously.

[0036] The plurality of plugs **40** are disposed to be in contact with either one or both of the second semiconductor region **22** or the third semiconductor region **23**. A plurality of regions formed in the Y direction may be collectively referred to as the plug **40**, but each of the plurality of regions formed may also be referred to as the plug **40**.

[0037] Lengths L1, L2, L3 and LG shown in FIG. 1B will be described in detail after FIG. 2 is described.

[0038] Hereinafter, a cross-sectional structure of the semiconductor device **1** according to the first embodiment will be described with reference to FIGS. 2A, 2B and 2C.

[0039] FIGS. 2A, 2B and 2C are cross-sectional views taken along lines A-A', B-B' and C-C' shown in FIG. 1B, respectively. The cross-sectional views of FIGS. 2A, 2B and 2C show the first electrode **11** and the interlayer insulating film **50**, which are omitted in FIG. 1B. Further, in the cross-sectional views of FIGS. 2B and 2C, a structure below the first semiconductor region **21** is omitted.

[0040] First, the A-A' cross-sectional view of the semiconductor device **1** according to the first embodiment will be described with reference to FIG. 2A. The semiconductor device **1** according to this embodiment shown in FIG. 2A includes the first electrode **11**, a second electrode **12**, the semiconductor substrate **20**, the gate region **30**, the plug **40**, and the interlayer insulating film **50**. The semiconductor substrate **20** includes the first semiconductor region **21** of a first conductivity type, the second semiconductor region **22** of a second conductivity type, the third semiconductor region **23** of a first conductivity type, a fourth semiconductor region **24** of a second conductivity type, and a fifth semiconductor region **25** of a second conductivity type. The gate region **30** includes a gate insulating film **31** and a gate electrode **32**.

[0041] The first electrode **11** and the second electrode are, for example, metals containing AlCu or AlSi. The first electrode **11** is, for example, an emitter electrode of an IGBT. The second electrode **12** is, for example, a collector electrode of an IGBT. The semiconductor substrate **20** is provided between the first electrode **11** and the second electrode **12**.

[0042] The semiconductor substrate **20** is made of, for example, a semiconductor containing Si. The first semiconductor region **21** is, for example, an n-type drift layer. The second semiconductor region **22** is, for example, a p-type base region. The third semiconductor region **23** is, for example, an n.sup.+ -type emitter layer. The fourth semiconductor region **24** is, for example, a p.sup.+ -type collector layer. The fifth semiconductor region **25** is, for example, a p.sup.+ -type contact region. The p-type semiconductor layer is formed, for example, by doping Si substrate with B. The n-type semiconductor layer is formed, for example, by doping Si substrate with N or P.

[0043] The gate region **30** is, for example, a gate trench of an IGBT. The gate insulating film **31** contains, for example, silicon oxide such as SiO₂. The gate insulating film **31** is, for example, a gate insulating film of an IGBT and is formed by oxidation. The gate electrode **32** is, for example, a gate electrode of an IGBT, and contains polysilicon.

[0044] It is preferable that the plug **40** be made of a metal containing W (tungsten). The plug **40** may also be covered with a barrier metal **41**. The barrier metal **41** is formed between the plug **40** and the semiconductor substrate **20** and between the plug **40** and the interlayer insulating film. The barrier metal **41** has a layer structure containing, for example, Ti and TiN. The barrier metal **41** has a higher adhesion to the interlayer insulating film **50** than the plug **40**, and prevents the plug **40** from peeling-off from the interlayer insulating film **50** when stress is applied to the semiconductor substrate **20** and the interlayer insulating film **50** due to temperature changes or the like. Further, if the plug **40** is formed on the surface of the barrier metal **41**, for example by CVD, the reactivity between a gas used for the CVD and the barrier metal **41** can be controlled by selecting the material of the barrier metal **41**. For example, when the CVD is performed using a gas containing fluorine, the reactivity of fluorine can be reduced by the barrier metal **41** containing TiN.

[0045] The plug **40** may be covered with the barrier metal **41**, and the barrier metal **41** may be in contact with the semiconductor substrate **20**. The plug **40** and the semiconductor substrate **20** are electrically connected to each other in a contact portion where the plug **40** and the semiconductor substrate **20** are in contact with each other. In the description of this specification, a contact area between the plug **40** and the semiconductor substrate **20** is described, but the definition of "contact" also includes a case where the plug **40** and the semiconductor substrate **20** face each other via the

barrier metal **41**.

[0046] The plug **40** electrically connects the first electrode **11** and the semiconductor substrate **20**. The plug **40** is in contact with the third semiconductor region **23** in the A-A' cross section and reaches the second semiconductor region **22**. The fifth semiconductor region **25** is formed on the bottom surface of the plug **40**.

[0047] The gate region **30** is provided below the interlayer insulating film **50** and reaches the first semiconductor region **21** in the negative direction of the Z direction. The gate region **30** includes a gate electrode **32** covered with the gate insulating film **31**. The gate insulating film **31** is provided between the gate electrode **32** and the semiconductor substrate **20**, and at least a portion of the gate insulating film **31** is in contact with the second semiconductor region **22**. The gate electrode **32** is electrically insulated from the semiconductor substrate **20** by the gate insulating film **31**. By applying a voltage to the gate electrode **32**, an inversion layer is formed in the second semiconductor region **22**, and the semiconductor device can be set to be in an ON state.

[0048] The interlayer insulating film **50** is made of, for example, silicon oxide such as SiO₂. The interlayer insulating film **50** is formed by, for example, CVD. The interlayer insulating film **50** is formed on the semiconductor substrate **20** and is provided to cover the gate region **30**. The gate electrode **32** and the first electrode **11** are electrically insulated from each other by the interlayer insulating film **50**.

[0049] FIG. **2B** is an enlarged cross-sectional view taken along a line B-B' shown in FIG. **1B**. Since the B-B' cross section does not pass through the plug **40**, the plug **40** is not shown in FIG. **2B**. On the other hand, the gate region **30** extends in the Y direction in FIG. **1B**, and thus the gate region **30** is shown in FIG. **2B** in the same manner as in FIG. **2A**.

[0050] FIG. **2C** is an enlarged cross-sectional view taken along a line C-C' shown in FIG. **1B**. A plurality of plugs **40** are provided in the Y direction. FIG. **2C** also shows a case where a plurality of third semiconductor regions **23** are provided in the Y direction. The arrangement of the plugs **40** in the Y direction can be changed depending on the arrangement of the third semiconductor regions **23** in the Y direction. The plugs **40** and the third semiconductor regions **23** are each provided at a fixed interval in the Y direction, for example.

[0051] Some of the plugs **40** may be in contact with the third semiconductor regions **23** and some may be located between the plurality of third semiconductor regions **23**. At least one plug **40** is provided between the third semiconductor regions **23** spaced apart in the Y direction. For example, as shown in FIG. **2C**, the plugs **40** in contact with the third semiconductor regions **23** and the plugs **40** located between the plurality of third semiconductor regions **23** are alternately repeated in the Y direction. As will be described below with reference to FIG. **2D**, the plugs **40** in contact with the third semiconductor regions **23** and the plugs **40** located between the plurality of third semiconductor regions **23** do not have to be alternately repeated in the Y direction. FIG. **2D** is a diagram showing an example different from the embodiment shown in FIG. **2C** with respect to the cross section in the same direction as in FIG. **2C**.

[0052] As shown in FIG. **2D**, in some of the plurality of plugs **40**, the plugs **40** adjacent to each other in the Y direction may be in contact with the third semiconductor region **23**. For example, two or more plugs **40** may be in contact with one region among the plurality of third semiconductor regions **23**.

[0053] Although not shown in FIG. **2D**, the plugs **40** adjacent to each other in the Y direction may be in contact with different third semiconductor regions **23** spaced apart from each other in the Y direction among the plurality of third semiconductor regions **23**. In other words, the third semiconductor region **23**, the second semiconductor region **22**, and the third semiconductor region **23** may be arranged in the Y direction between the plugs **40** adjacent to each other in the Y direction.

[0054] Alternatively, as shown in FIG. **2D**, for at least some of the plurality of plugs **40**, the plugs **40** adjacent to each other in the Y direction may be located between the third semiconductor

regions **23** that are spaced apart from each other in the Y direction. Two or more plugs **40** may be provided between the third semiconductor regions **23** adjacent to each other in the Y direction among the plurality of third semiconductor regions **23**.

[0055] The arrangement of the plugs **40** of the semiconductor device **1** according to the first embodiment will be further described below with reference to FIG. **1B** and FIGS. **2A**, **2B**, and **2C**.

[0056] First, description is given with reference to FIG. **1B**. As shown in FIG. **1B**, when comparing an interval LG between the gate regions **30** in the X direction and a length L1 of the plug **40** in the X direction, L1 is shorter. By making L1 smaller than LG, the plug **40** can be formed between the gate regions **30**. L1 (<LG) cannot necessarily be set to any length smaller than LG, and may be set to be sufficiently smaller than LG in consideration of an error of alignment in the X direction when the plug **40** is formed.

[0057] In addition, the length L1 of the plug **40** in the X direction and the length L2 in the Y direction are different from each other. L1 is the length of the long side of the plug **40**, for example, having a rectangular cross section in an XY plane, and L2 is the length of the short side of the plug **40**, for example, having a rectangular cross section in the XY plane.

[0058] An interval L3 between the plugs **40** adjacent to each other in the Y direction may be smaller or larger than L1. The plugs **40** can be provided at a fixed interval L3 as shown in FIG. **1B**.

[0059] The plug **40** is in contact with the second semiconductor region **22** or the third semiconductor region **23** in the XY plane shown in FIG. **1B**. At least a portion of the plug **40** may be in contact with both the second semiconductor region **22** and the third semiconductor region **23** in the XY plane.

[0060] Next, description is given with reference to FIG. **2A**. The plug **40** has a depth D in the Z direction. The depth D is the length in the Z direction from an interface between the plug **40** and the first electrode **11** to the bottom surface of the plug **40**. The shape of the bottom surface of the plug **40** is not limited to that shown in FIG. **2A**, and the bottom surface may be rounded.

[0061] A ratio between the depth D and the length L1, that is, $D/L1$, is referred to as a first aspect ratio. In addition, a ratio between the depth D shown in FIG. **2C** and the length L2 of the plug **40** in the Y direction, that is, $D/L2$, is referred to as a second aspect ratio.

[0062] The first aspect ratio $D/L1$ is, for example, within the range of $0 < D/L1 \leq 2$. If the first aspect ratio satisfies $0 < D/L1 \leq 2$, the embeddability when forming the plug **40** can be improved. It is preferable that the first aspect ratio be within the range of $0 < D/L1 \leq 1$.

[0063] The second aspect ratio $D/L2$ satisfies $D/L1 \leq D/L2$, for example, $1 \leq D/L2$. By making the second aspect ratio $D/L2$ satisfy $1 \leq D/L2$, a contact area between the plug **40** and the semiconductor substrate **20** can be increased, which is advantageous for miniaturization. Since the first aspect ratio is smaller than the second aspect ratio, satisfactory embeddability of the plug **40** can be maintained even if the second aspect ratio is large.

[0064] For example, the plug **40** may have a structure tapered in the negative direction of the Z direction. In this case, the lengths L1 and L2 may be defined as the maximum length in the X direction of a portion of the plug **40** which is located between the gate regions **30**. For example, in the Z direction, the length of the plug **40** in the X direction at a position corresponding to an interface between the semiconductor substrate **20** and the interlayer insulating film **50** is measured and set as L1.

[0065] Next, operations of the semiconductor device **1** will be described with reference to FIG. **2A**. As an example, operations of an IGBT will be described. Description will be given of an example in which the first electrode **11** is an emitter electrode, the first semiconductor region **21** is an n.sup.-type drift region, the second semiconductor region **22** is a p-type base region, the third semiconductor region **23** is an n.sup.+type emitter layer, the fourth semiconductor region **24** is a p.sup.+type collector layer, and the fifth semiconductor region **25** is a p.sup.+type contact region. First, an operation of turning on the IGBT will be described.

[0066] A negative voltage is applied to the first electrode **11** with respect to the second electrode

12. By applying a positive voltage to the gate electrode **32** which is equal to or greater than a threshold voltage with respect to the potential of the first electrode **11**, an inversion layer (channel) is formed in the second semiconductor region **22** in the vicinity of the gate insulating film **31**. That is, an inversion layer with n-type conductivity is formed in the second semiconductor region **22** which is a p-type base region.

[0067] The inversion layer formed in the second semiconductor region **22** has the same conductivity type as those of the first semiconductor region **21** and the third semiconductor region **23**. Thus, electrons that reach the third semiconductor region **23** from the first electrode **11** (emitter electrode) through the plug **40** flow to the first semiconductor region **21** through the inversion layer formed in the second semiconductor region **22**.

[0068] Furthermore, holes are injected into the first semiconductor region **21** (drift layer) from the fourth semiconductor region **24** (collector layer) provided below the first semiconductor region **21**. In this manner, a current flows from the second electrode **12** (collector electrode) to the first electrode (emitter electrode) with the electrons and the holes as current carriers, and the IGBT is set to be in an ON state.

[0069] Next, when the IGBT is turned off, the potential of the gate electrode **32** is set to, for example, the same potential as the first electrode **11**. The inversion layer formed in the second semiconductor region **22** is removed, and the injection of electrons from the third semiconductor region **23** to the first semiconductor region **21** is prevented. In order for the IGBT to be completely turned off, it is necessary to discharge carriers accumulated in the first semiconductor region **21** in an ON state. The electrons are discharged to the second electrode **12**, and the holes are discharged to the first electrode **11** through the p-type second semiconductor region **22** and the plug **40**.

[0070] The holes are discharged, for example, through the following route. The holes accumulated in the first semiconductor region **21** are discharged to the plug **40** through the second semiconductor region **22** and the fifth semiconductor region **25** which has the same conductivity type as the second semiconductor region. That is, the fifth semiconductor region **25** connects the plug **40** and the second semiconductor region **22** and is further in ohmic contact with the plug **40** to become a hole discharge route.

[0071] That is, the plug **40** is a route for flowing electrons from the first electrode **11** to the third semiconductor region **23** when the IGBT is in an ON state, and discharging holes from the second semiconductor region **22** to the first electrode **11** when the IGBT is turned off. As described above, the plug **40** electrically connects the first electrode **11** and the semiconductor substrate **20**, and the semiconductor device **1** operates.

[0072] With the semiconductor device **1** of this embodiment, each of the plurality of plugs **40** provided in the Y direction has the length L1 in the X direction and the length L2 in the Y direction different from L1, thereby making it possible to prevent defective formation of the plugs **40** and improve the reliability of the semiconductor device. By improving the flatness of the first electrode **11** formed on the plugs **40**, concentration of stress on a portion of the electrode and occurrence of cracks are prevented.

[0073] The plug **40** and the semiconductor substrate **20** are generally made of materials having different thermal expansion coefficients. For example, W (tungsten) contained in the plug **40** has a larger thermal expansion coefficient than that of Si (silicon) contained in the semiconductor substrate. When heat is applied, the plug **40** and the semiconductor substrate **20** differ in the degree of expansion, and thus distortion and stress may occur. Stress generated between the plug **40** and the semiconductor substrate **20** may cause warpage in a wafer. In general, as the width of the plug **40** increases, it is necessary to increase the film thickness of W (tungsten), and thus stress is increased, which results in a concern that the warpage of the wafer may become larger. That is, an increase in the size of a contact area between the plug **40** and the semiconductor substrate **20** may result in a concern that stress to be applied to a wafer may be increased.

[0074] Hereinafter, as an example, a case where the plug **40** is made of W (tungsten) will be

described. The plug **40** is formed, for example, by embedding a conductive material in an opening, which is formed by excavating the semiconductor substrate **20** and the interlayer insulating film **50** in the Z direction. A film containing a conductive material (for example, W) is embedded in the opening by being formed to have a predetermined thickness on the bottom surface and side walls of the opening. For example, in order to form the plug **40** with the length L_2 in the Y direction, a film with a thickness of $L_2/2$ is formed on the side walls. In order to fill the opening, a film with a thickness of at least $L_2/2$ is formed on the side walls of the opening. In order to satisfactorily fill the opening without impairing flatness and preventing the formation of voids, a film thickness is preferably $L_2 \times (3/5)$ or more. More preferably, the film thickness is $L_2 \times (3/4)$ or more.

[0075] Here, a length T_{max} is defined. The length T_{max} is a maximum value of a film thickness of W (tungsten) formed on the sidewall of the opening, the film thickness being a length at which the magnitude of stress on the wafer is equal to or less than an allowable predetermined value. The predetermined value allowable for stress is determined such that the amount of wafer warpage caused by the stress is sufficiently small. By setting the film thickness of the W (tungsten) formed on the sidewall of the opening to equal to or less than T_{max} , the reliability of the semiconductor device is improved.

[0076] In order to prevent the wafer warpage, a necessary thickness $L_2/2$ of the film formed on the sidewall of the opening satisfies $L_2/2 \leq T_{max}$. In other words, $L_2 \leq 2T_{max}$. The film with a thickness of $L_2/2$ which is formed on the sidewall of the opening has a film thickness of T_{max} or less, and thus the stress on the wafer can be reduced. That is, the length L_2 of the plug **40** in the Y direction needs to be set to a length of $2T_{max}$ or less in order to reduce the stress on the wafer.

[0077] In order to fill the opening more satisfactorily, it is preferable that $L_2 \leq (5/3) T_{max}$. The film with a thickness of $(3/5) L_2$ which is formed on the side wall of the opening has a film thickness of T_{max} or less, and thus the stress on the wafer can be reduced. It is further preferable that $L_2 \leq (4/3) T_{max}$.

[0078] For example, the plug **40** is formed such that $L_2 \leq (4/3) T_{max}$, and thus L_1 can be arbitrarily increased within a range without exceeding LG . This is because the W (tungsten) film is formed to be thin in the Y direction in order to prevent wafer warpage. Furthermore, the opening can be filled satisfactorily by preventing the formation of voids. In other words, it is possible to prevent defective formation of the plug **40**.

[0079] Since L_1 may be determined without considering the influence of the film thickness of W on the wafer warpage, the length of a region where the plug **40** is in contact with the second semiconductor region **22** or the third semiconductor region **23** can be increased in the X direction to increase a contact area. Thus, on-resistance is reduced, and the discharge of carriers from the semiconductor substrate **20** to the plug **40** is promoted during switching, whereby a switching loss can be reduced. By reducing a switching loss, power consumption is reduced, and it is possible to provide the semiconductor device with improved switching performance.

[0080] Since L_1 may be determined without considering the influence of the film thickness of W on the wafer warpage, according to this embodiment, the plug **40** with satisfactory embeddability can be formed when the range of the value of the interval LG between the gate regions **30** in the X direction is wide. The wide range of the value of the interval LG will be described below in comparison with first and second reference examples.

[0081] In addition, according to this embodiment, the plurality of third semiconductor regions **23** are provided in the Y direction, and plugs may be formed in contact with the third semiconductor regions **23**, or plugs **40** may be formed in contact with the second semiconductor region **22** between the third semiconductor regions **23** spaced apart from each other in the Y direction. A current density in the XY plane can also be controlled by changing either a layout of the third semiconductor region **23** or a layout of the plugs **40**. By increasing the degree of freedom in design and controlling the amount of heat generated by a current, the reliability of the semiconductor device can be improved.

[0082] The third semiconductor region **23** is a region serving as a path for a current when the semiconductor device **1** is in an ON state. The shorter a distance between the plurality of plugs **40** in contact with the third semiconductor region **23** becomes, in other words, as the degree of dispersion of the plurality of plugs **40** in contact with the third semiconductor region **23** in the XY plane becomes lower, the higher the current density in an ON state. In contrast, by increasing the degree of dispersion of the plurality of plugs **40** in contact with the third semiconductor region **23** in the XY plane, it is possible to reduce the current density and reduce the amount of heat locally generated.

[0083] For example, the plugs **40** adjacent to each other in the Y direction may be located between the third semiconductor regions **23** that are spaced apart from each other in the Y direction. That is, it is possible to reduce the current density and the amount of heat generated by increasing the proportion of the plugs **40** in contact with the second semiconductor region **22**. Furthermore, a wide carrier discharge path can be formed by increasing a contact area between the second semiconductor region **22** and the plugs **40**, thereby reducing a switching loss.

[0084] After the amount of heat generated is reduced, for example, the plugs **40** adjacent to each other in the Y direction can be allowed to be in contact with the third semiconductor region **23** at a portion, thereby increasing the current density. In a region with a relatively high heat dissipation, the current density can be designed to be high.

[0085] In addition, some of the plugs **40** may be provided to straddle the second semiconductor region **22** and the third semiconductor region **23** in the Y direction. By changing a positional relationship between the third semiconductor region **23** and the plugs **40**, the degree of dispersion of the plurality of plugs **40** in contact with the third semiconductor region **23** in the XY plane can be adjusted. That is, by adjusting the arrangement of the third semiconductor regions **23** and the arrangement of the plugs **40**, the amount of heat generated and the current density can be optimized in accordance with the electrical characteristics of the semiconductor device.

[0086] The degree of dispersion of the plurality of plugs **40** in contact with the third semiconductor region **23** in the XY plane can be changed in the XY plane. For example, the degree of dispersion may be increased toward the end of the active region AR shown in FIG. **1**. It is possible to reduce the amount of heat generated at the end of the active region AR and improve reliability.

[0087] According to this embodiment, it is possible to control the current density in the XY plane by the layout of the plugs **40** without necessarily changing the layout of the third semiconductor region **23**. The degree of freedom in design for controlling the current density is improved. Furthermore, it is possible to reduce the current density by locally changing a positional relationship between the third semiconductor region **23** and the plugs **40** in a region in the XY plane where the amount of heat generated is desired to be reduced. At the same time, the layout of the third semiconductor region **23** can be changed to adjust the layout of the third semiconductor region **23** and the layout of the plugs **40**, thereby further increasing the degree of freedom in design.

[0088] Furthermore, as compared to a case where a current is controlled by changing only the layout of the third semiconductor region **23**, it is possible to expand the range of selection according to the arrangement of the plugs **40** for a predetermined layout of the third semiconductor region **23**. By appropriately selecting the arrangement of the plugs **40**, the electrical characteristics of the semiconductor device can be improved. That is, current flow can be further controlled by the arrangement of the plugs **40** in the X and Y directions, in addition to the layout of the third semiconductor region **23**.

Modification of First Embodiment

[0089] Next, the semiconductor device **1** according to a modification of the first embodiment will be described with reference to FIG. **3**. FIG. **3** is an enlarged plan view of the semiconductor device **1** according to the modification of the first embodiment. A difference between FIG. **3** and FIG. **1B** is in a cross-sectional shape of the plug **40** in the XY plane. FIG. **3** shows a case where the plug **40**

has an elliptical shape having a length L1 in the X direction and a length L2 in the Y direction. The length L1 in the X direction is the length of the major axis of the ellipse, and the length L2 in the Y direction is the length of the minor axis of the ellipse. For example, $L2 < L1$, and $L2 \leq 2T_{max}$. A relationship of $L2 \leq (5/3) T_{max}$ or $L2 \leq (4/3) T_{max}$ may be established.

[0090] In addition, the cross-sectional shape of the plug **40** in the XY plane is not limited to the rectangular shape shown in FIG. 1B or the elliptical shape shown in FIG. 3. For example, the cross-sectional shape may be an oval shape or a polygonal shape, or may be a polygonal shape with rounded corners.

[0091] More generally, any point inside a figure (for example, a rectangle) representing the cross-sectional shape of the plug **40** in the XY plane may be located within a distance of, for example, T_{max} from at least one point on the outer periphery of the figure. The cross-sectional shape of the plug **40** in the XY plane in at least one direction has a length of $2T_{max}$ or less. When the plug **40** is formed, the opening can be satisfactorily filled with W (tungsten) having a film thickness of T_{max} or less.

[0092] For example, in the elliptical shape shown in FIG. 3, the length of the minor axis $L2 \leq 2T_{max}$. A distance between the center of the ellipse and a point on the circumference of the ellipse in both directions in the Y direction (minor axis direction) is equal to or less than T_{max} . That is, the center of the ellipse can be satisfactorily filled with a tungsten film having a film thickness of T_{max} or less from the sidewall of the opening. The same is true of any point inside the ellipse, which is the cross-sectional shape of the plug **40** shown in FIG. 3.

[0093] In addition, the cross-sectional shape of the plug **40** in the XY plane may be curved and may be, for example, an L-shape. Alternatively, the cross-sectional shape may be a cruciform.

[0094] That is, the shape of the plug **40** is not limited to an elliptical shape, but may be any shape that can prevent defective formation of the plug **40**. A figure with long and short sides, such as a rectangle, is preferable because it is possible to prevent defective formation of the plug **40** on the short side and take up a large length of a region where the plug **40** and the semiconductor substrate **20** are in contact with each other on the long side. In this specification, a shape with long and short sides is defined as a "rectangle."

[0095] The present disclosure is not limited to the semiconductor device **1** according to the first embodiment shown in FIG. 1B, and it is possible to prevent defective formation of the plug **40** and improve the reliability of the semiconductor device by, for example, the modification of the first embodiment shown in FIG. 3. At the same time, it is possible to reduce a switching loss and improve the switching performance of the semiconductor device by taking a wide contact area between the plug **40** and the second semiconductor region **22**. The modification of the first embodiment is described above.

First Reference Example

[0096] Next, a semiconductor device **101** according to a first reference example will be described with reference to FIGS. 4 and 5. Differences from the first embodiment will be described in detail. FIGS. 4 and 5 are an enlarged plan view and a cross-sectional view of the semiconductor device **101** according to the first reference example, respectively. FIG. 4 is an enlarged plan view showing an internal wiring structure, and FIG. 5 is a cross-sectional view showing a D-D' cross section of the semiconductor device shown in FIG. 4.

[0097] First, a plan view of the first reference example will be described with reference to FIG. 4. In the semiconductor device **101**, plugs **140** are provided in two rows in the X direction and extend in the Y direction. The plurality of plugs **140** are provided between a plurality of gate regions **130** provided in the X direction and extending in the Y direction. The plugs **140** are in contact with a second semiconductor region **122** and a third semiconductor region **123** of a semiconductor substrate **120**.

[0098] The gate region **130** includes a gate electrode **132** and a gate insulating film **131** that covers the gate electrode **132**. The gate electrode **132** and the gate insulating film **131** extend in the Y

direction. The gate electrode **132** is electrically insulated from the semiconductor substrate **120**.

[0099] The second semiconductor region **122** and the third semiconductor region **123** are semiconductor regions having different conductivity types. For example, the second semiconductor region **122** is a p-type base region, and the third semiconductor region **123** is an n-type emitter region.

[0100] The width in the X direction of the plug **140** in the semiconductor device **101** according to the first reference example is referred to as L4, and an interval in the X direction between the plugs **140** arranged in two rows in the X direction is referred to as L5. In addition, an interval in the X direction between the gate regions **130** adjacent to each other in the X direction is referred to as LG.

[0101] Next, a D-D' cross-sectional view of the semiconductor device **101** according to the first reference example will be described with reference to FIG. 5. The plug **140** is provided from a first electrode **111** to the semiconductor substrate **120** through an interlayer insulating film **150**. In the D-D' cross section shown in FIG. 5, the plug **140** is in contact with the third semiconductor region **123**.

[0102] The gate region **130** is filled with the gate electrode **132** and the gate insulating film **131** that covers the gate electrode **132**. The gate insulating film **131** reaches a first semiconductor region **121** of the semiconductor substrate **120**. The gate insulating film **131** is in contact with the second semiconductor region **122**.

[0103] In FIGS. 4 and 5, $2L4+L5 \leq LG$, and two rows of plugs **140** are provided in the X direction. If $2L4+L5 > LG$, two rows of plugs **140** cannot be formed between the gate regions **130**. In other words, an opening for embedding two rows of plugs **140** cannot be formed between the gate regions **130**.

[0104] The opening is formed by excavating the semiconductor substrate **120** and the interlayer insulating film **150** in order to embed the plugs **140**. The plugs **140** are formed by forming a W (tungsten) film in the opening. In order to form the plugs **140** shown in FIG. 5, it is necessary to form two rows of openings with a width of L4 in the X direction at an interval of L5 in the X direction.

[0105] In general, L4 cannot be made infinitely small. L4 cannot be set to be equal to or less than a length Lmin which is determined from the precision of excavation of the semiconductor substrate, and $L4 \geq Lmin$. When an opening for embedding W (tungsten) is formed, there is a limit to the processing accuracy for excavating the semiconductor substrate **120** and the interlayer insulating film **150**. That is, the processing accuracy in the excavation process may determine the minimum value of L4 that can be realized. In this manner, the length Lmin can be defined as the minimum value of L4 determined by the constraints of the processing accuracy. That is, Lmin can be defined as the minimum diameter of the opening to be provided in the semiconductor substrate.

[0106] Similarly, L5 also has a limit due to the processing accuracy, and the minimum value of L5 that can be realized can also be determined by the processing accuracy. For example, the inequality $L5 \geq Lmin$ holds because the processing accuracy for forming the opening by excavation and the processing accuracy for forming a region (mesa portion) that remains unexcavated are substantially the same. Hereinafter, a discussion will be made assuming that both the minimum values of L4 and L5 are Lmin.

[0107] If $LG < 3Lmin$, $2L4+L5 (\geq 3Lmin) > LG$ holds due to the limit of the processing accuracy at the forming the opening, so it is difficult to form two rows of plugs **140** in the X direction as shown in FIGS. 4 and 5. That is, with the semiconductor device **101** of the first reference example, If $LG < 3Lmin$, it is difficult to form the plugs **140**.

[0108] In addition, according to the first reference example, If LG is large, it may not be possible to increase the length in the X direction of the region where the plugs **140** and the semiconductor substrate **120** are in contact with each other. This is because, if $L4 > 2Tmax$, a thickness of at least Tmax or more is required as the film thickness of W in order to avoid the generation of voids,

which increases stress on the wafer and promotes wafer warpage. In order to secure the reliability of the semiconductor device, it is preferable to set the length L_4 to be in the range of $L_4 \leq 2T_{\max}$. In order to secure flatness, it is preferable to set the length to be in the range of $L_4 \leq (4/3) T_{\max}$. For simplicity, the case of $L_4 \leq 2T_{\max}$ will be described below, but the same is true of the cases of $L_4 \leq (5/3) T_{\max}$ and $L_4 \leq (4/3) T_{\max}$.

[0109] If two rows of plugs **140** are formed in the X direction and the reliability of the semiconductor device is secured, the length in the X direction of the region where the plugs **140** and the semiconductor substrate **120** are in contact with each other cannot be greater than $4T_{\max}$. That is, the length L_4 cannot necessarily be set to be large in proportion to the interval LG between the gate regions **130** in the X direction, and it is difficult to increase a contact area between the plugs **140** and the semiconductor substrate **120** in accordance with the value of LG .

[0110] Specifically, If the range of the value of LG is $4T_{\max} + L_{\min} < LG$, it becomes difficult to increase the contact area between the plug **140** and the semiconductor substrate **120** in the first reference example. This is because L_5 requires a minimum length of $L_{\min}$, and even if the value of LG is increased beyond $4T_{\max} + L_{\min}$, each of the two rows of plugs **140** can only have a maximum width of $2T_{\max}$.

[0111] In order to prevent an increase in stress and wafer warpage if LG is increased in the range of $LG > 4T_{\max} + L_{\min}$, L_4 is limited to a length that satisfies $L_4 \leq 2T_{\max}$, and the length of a portion that does not form the plug **140** (the length of L_5 or the length in the X direction between the plug **140** and the gate region **130**) is increased. Thus, it is difficult to increase the contact area between the plugs **140** and the semiconductor substrate **120** in accordance with the value of LG . Since it is difficult to increase the contact area between the plugs **140** and the semiconductor substrate **120**, it is difficult to promote the discharge of carriers in a turn-off state.

[0112] In summary, with the semiconductor device **101** of the first reference example, if $LG < 3L_{\min}$, it is difficult to form the plugs **140** between the gate regions **130**. In addition, if $4T_{\max} + L_{\min} < LG$, it is difficult to increase the contact area between the plugs **140** and the semiconductor substrate **120** as LG increases beyond $4T_{\max} + L_{\min}$. That is, it can be said that the structure in the first reference example is suitable only for LG within the range of $3L_{\min} \leq LG \leq 4T_{\max} + L_{\min}$.

[0113] On the other hand, with the semiconductor device **1** of the first embodiment, the plugs **40** can be formed regardless of the interval LG between the gate region **30** in the X direction. In particular, even if $(L_{\min} <) LG < 3L_{\min}$, the plugs **40** can be formed in the semiconductor device **1**.

[0114] Furthermore, with the semiconductor device **1** of the first embodiment, if LG is large, L_1 can also be increased in proportion to LG , and the contact area between the plugs **40** and the semiconductor substrate **20** can be increased in accordance with the value of LG . According to the semiconductor device **1** of the first embodiment, it is possible to increase the contact area between the plugs **40** and the semiconductor substrate **20** in accordance with LG , even in the range of $4T_{\max} + L_{\min} < LG$, in which it becomes difficult in the first reference example to increase the contact area between the plugs **140** and the semiconductor substrate **20**. Thus, it is possible to promote the discharge of carriers in a turn-off state and reduce a switching loss.

[0115] With the semiconductor device **1** of the first embodiment, it is possible to improve the reliability and performance of the semiconductor device for a wider range of LG than in the first reference example, that is, for more diversified arrangement of the gate regions **30**.

Second Reference Example

[0116] Next, a semiconductor device **102** according to a second reference example will be described with reference to FIG. 6.

[0117] FIG. 6 is an enlarged plan view showing the semiconductor device **102** according to the second reference example. The semiconductor device **102** according to the second reference example is different from the semiconductor device **101** according to the first reference example in that the plugs **140** are provided in a row in the X direction and extend in the Y direction. The width

in the X direction of the plugs **140** provided in a row in the X direction is referred to as L6. As in the first reference example, an interval between the gate regions **130** in the X direction is referred to as LG.

[0118] With the semiconductor device **102** of the second reference example, it is possible to form the plugs **140** if L6 is smaller than or equal to LG. In other words, using a length Lmin determined by the accuracy of excavation for the length L6, it is possible to form the plugs **140** if $LG \geq Lmin$.

[0119] On the other hand, if $L6 > 2Tmax$, it is difficult to prevent defective formation of the plugs **140** while preventing wafer warpage. That is, in order to secure the reliability of the semiconductor device, it is preferable that L6 have a length that satisfies $L6 \leq 2Tmax$.

[0120] In the semiconductor device **102** according to the second reference example, it is necessary to form the plugs **140** in a range that satisfies $L6 \leq 2Tmax$ in order not to impair the reliability of the semiconductor device. As LG increases in the range of $LG > 2Tmax$, it becomes difficult to increase a contact area between the plugs **140** and the semiconductor substrate **120** in accordance with LG.

[0121] In summary, with the semiconductor device **102** of the second reference example, only for LG within the range of $Lmin \leq LG \leq 2Tmax$, the contact area between the plugs **140** and the semiconductor substrate **120** can be increased to promote the discharge of carriers and improve the performance of the semiconductor device.

[0122] On the other hand, with the semiconductor device **1** of the first embodiment, L1 is set to be large in accordance with LG, and thus the contact area between the plugs **40** and the semiconductor substrate **20** can be increased even for LG in the range of $2Tmax < LG$ to promote the discharge of carriers and improve the switching performance of the semiconductor device.

[0123] The semiconductor device **102** according to the second reference example is described above.

[0124] Next, the semiconductor device **101** according to the first reference example and the semiconductor device **102** according to the second reference example will be described in comparison with the semiconductor device **1** according to the first embodiment. To summarize, the second reference example is not suitable if LG is in the range of $LG < Lmin$ or $2Tmax < LG$, and the first reference example is not suitable if LG is in the range of $LG < 3Lmin$ or $4Tmax + Lmin < LG$. Combining these two inequalities and assuming $2Tmax < 3Lmin$ for simplicity, if LG is set to be in the range of $LG < Lmin$ (from the second reference example), $2Tmax < LG < 3Lmin$ ($2Tmax < LG$ from the second reference example and $LG < 3Lmin$ from the first reference example) or $4Tmax + Lmin < LG$ (from the first reference example), neither the structure in the first reference example nor the structure in the second reference example is suitable.

[0125] On the other hand, in the semiconductor device **1** according to the first embodiment, the contact area between the plugs **140** and the semiconductor substrate **20** can be increased for any value of LG greater than Lmin.

[0126] In both the semiconductor device **101** according to the first reference example and the semiconductor device **102** according to the second reference example, it becomes difficult to improve the performance if the interval LG between the gate regions **130** in the X direction is within a specific range.

[0127] That is, depending on the value of LG, it is necessary to select each time which of the semiconductor device **101** according to the first reference example or the semiconductor device **102** according to the second reference example is applied. Furthermore, there is a problem that there is a range of LG ($2Tmax < LG < 3Lmin$ or $4Tmax + Lmin < LG$) in which the plugs **140** cannot be efficiently formed with any of the semiconductor device **101** according to the first reference example or the semiconductor device **102** according to the second reference example.

[0128] Here, "efficiently forming the plugs **140**" means that it is possible to increase the contact area between the plugs **140** and the semiconductor substrate **20** in accordance with LG as LG increases. In contrast, when it is difficult to increase the contact area between the plugs **140** and the semiconductor substrate **20** in response to an increase in LG, it can be said that the plugs **140**

cannot be efficiently formed.

[0129] The length LG may be determined by the configuration of the semiconductor device. For example, an interval between gate regions of an IGBT may be determined based on characteristics required for the semiconductor device, such as a breakdown voltage. Thus, LG cannot necessarily be selected arbitrarily, and there may be cases where a gate region **30** has to be formed such that LG has a value within the range of $2T_{\max} < LG < 3L_{\min}$ or $4T_{\max} + L_{\min} < LG$.

[0130] If the LG value required for the semiconductor device is within the range of $2T_{\max} < LG < 3L_{\min}$ or $4T_{\max} + L_{\min} < LG$, neither the first nor the second reference example is appropriate. That is, neither the first nor the second reference example can efficiently form the plug **140**.

[0131] Furthermore, for example, when focusing on the range of the value of LG, that is, $3L_{\min} \leq LG \leq 4T_{\max} + L_{\min}$, for which it can be determined that the first reference example is suitable, the range of the value of LG satisfying the inequality is limited as a difference between $2T_{\max}$ and $L_{\min}$ becomes smaller. That is, if $2T_{\max}$ and $L_{\min}$ become substantially the same length, the range of LG for which the structure in the first reference example is suitable becomes narrower. The same is true of the range of the value of LG for which the second reference example is suitable, that is, $L_{\min} < LG < 2T_{\max}$.

[0132] That is, for example, the more it becomes necessary to reduce the film thickness of a tungsten film by reducing stress on a wafer, the more limited the range of the value of LG for efficiently forming the plugs **140** becomes according to the first and second reference examples. As the range of the value of LG becomes more limited, the degree of freedom in designing the semiconductor device decreases.

[0133] The inequality for LG is described above. On the other hand, it may be necessary to consider the inequality for L1 as well. The inequality for L1 can be considered when L1 is set to be smaller than LG by a certain value as a measure to deal with an error of alignment in the X direction at the time of forming the plugs **40**. That is, in reality, L1 is not necessarily set to a length equal to LG (even if $LG > L_{\min}$, the plug **40** cannot necessarily be formed), and it may be appropriate to consider an inequality for the length L1.

[0134] For example, if $L_{\min} < LG$, the plug **40** shown in FIG. **1B** cannot be necessarily formed. This is because there is a concern that the plug **40** and the gate electrode **32** will be electrically connected due to an error of alignment in the X direction at the time of forming the plug **40** having L1 which is a length equal to LG. Consequently, it is considered that L1 is set to be smaller than LG by the error of the alignment so that the plug **40** and the gate electrode **32** are not electrically connected. In this case, even if $L_{\min} < LG$, there is a concern that $L1 < L_{\min}$, which eliminates realizability in consideration of the accuracy of excavation.

[0135] On the other hand, if $L_{\min} < L1$, the plug **40** shown in FIG. **1B** can be formed from the viewpoint of the accuracy of excavation. That is, considering an error of alignment, $L_{\min} < LG$ and $L_{\min} < L1$ do not necessarily have the same meaning. Thus, it is preferable to consider the inequality for LG and the inequality for L1.

[0136] In the semiconductor device **1** according to the first embodiment shown in FIG. **3**, the inequality for LG can be rephrased as the inequality for L1. For example, the inequality $2T_{\max} < LG < 3L_{\min}$ can be rephrased as an inequality for L1 ($2T_{\max} < L1 < 3L_{\min}$). When the inequality for LG is rephrased as the inequality for L1, the semiconductor device **1** according to the first embodiment can form the plug **40** even if $2T_{\max} < L1 < 3L_{\min}$ or $4T_{\max} + L_{\min} < L1$ ($< 5L_{\min}$).

[0137] Hereinafter, more generally, a case where n rows (n is a natural number) of plugs **140** are formed in the X direction between the gate regions **130** (not shown) will be described. The first reference example corresponds to a case where $n=2$, and the second reference example corresponds to a case where $n=1$.

[0138] For example, if $n=3$, three rows of plugs **140** with a width of L4 are formed in the X direction, and the semiconductor substrate **20** with a width of L5 is provided between the plugs

140. Thus, since a relationship of $3L_4 + 2L_5 \leq LG$ has to be satisfied, and $L_{min} \leq L_4$ and $L_{min} \leq L_5$, a relationship of $5L_{min} \leq LG$ needs to be satisfied to form three rows of plugs **140**. If three rows of plugs **140** are formed in the X direction, it becomes difficult to form three rows of plugs **140** in the X direction for LG in the range of $LG < 5L_{min}$.

[0139] In addition, if $n=3$ and $6T_{max} + 2L_{min} < LG$, it becomes difficult to increase a contact area between the plugs **140** and the semiconductor substrate **20** in proportion to LG, that is, to form the plugs **140** efficiently. In other words, if $n=3$, the plugs **140** can be formed efficiently only within the range of $5L_{min} \leq LG \leq 6T_{max} + 2L_{min}$.

[0140] Thus, the range of the value of LG ($4T_{max} + L_{min} < LG < 5L_{min}$) exists between the range of the value of LG for which the second reference example ($n=2$) is appropriate, that is, $3L_{min} \leq LG \leq 4T_{max} + L_{min}$, and the range of the value of LG for which an example of $n=3$ is appropriate.

[0141] By a similar discussion, in general, if n rows of plugs **140** (n is a natural number) are formed in the X direction between the gate regions **130**, the plugs **140** cannot be formed efficiently unless $(2n-1) \times L_{min} \leq LG \leq n \times 2T_{max} + (n-1) \times L_{min}$.

[0142] As described above, in the case of LG in the range of $2T_{max} < LG < 3L_{min}$, the plugs **140** cannot be formed efficiently by either the second reference example ($n=1$) or the first reference example ($n=2$). Similarly, when LG is in the range of an inequality of $n \times 2T_{max} + (n-1) \times L_{min} < LG < (2n+1) \times L_{min}$ for a certain natural number n , it is difficult to efficiently form the n rows of plugs **140** in the X direction.

[0143] That is, in a method of forming the n rows of plugs **140** in the X direction, manufacturing costs may increase due to changing the number of rows of plugs **140** formed in the X direction depending on the value of LG.

[0144] Furthermore, as is apparent from an inequality of $(2n-1) \times L_{min} < LG < n \times 2T_{max} + (n-1) \times L_{min}$, for any natural number n , the range of the value of LG for which it is appropriate to form n rows of plugs **140** becomes narrow if the film thickness of the tungsten film has to be reduced for reducing stress on the wafer.

[0145] On the other hand, with the semiconductor device **1** of the first embodiment, L_1 can be set to be large in accordance with LG regardless of the value of LG, and thus a contact area between the plugs **40** and the semiconductor substrate **20** can be increased. That is, the plugs **40** can be formed efficiently for a wider range of the value of LG. Specifically, even if LG is within the range of the inequality $n \times 2T_{max} + (n-1) \times L_{min} < LG < (2n+1) \times L_{min}$ for a certain natural number n , the plugs **40** can be formed efficiently. Thus, the degree of freedom in designing the semiconductor device is increased as compared to a method of forming n rows of plugs **140** in the X direction.

[0146] With the semiconductor device **1** of the first embodiment, it is possible to efficiently form the plugs **40** even if the range of the value of LG is wider. For example, even if the value of LG is restricted due to the electrical characteristics of the semiconductor device, the plugs **40** can be efficiently formed. By increasing the range of selection of the value of LG, it is possible to increase the range of selection according to optimization of current control, for example, when the magnitude and distribution of a current flowing through the semiconductor device are controlled in accordance with the arrangement of the gate region **30**. It is possible to improve the electrical characteristics by appropriately selecting the value of LG.

Second Embodiment

[0147] Next, a semiconductor device **2** according to a second embodiment will be described with reference to FIG. 7 and FIGS. 8A and 8B. The semiconductor device **2** according to the second embodiment is an example of a structure that can be implemented by increasing the degree of freedom in designing a third semiconductor region **23** and a plug **40**.

[0148] FIG. 7 and FIGS. 8A and 8B are an enlarged plan view and cross-sectional views showing the semiconductor device **2** according to the second embodiment, respectively. FIG. 7 is an enlarged plan view, FIG. 8A shows an E-E' cross section of the semiconductor device shown in

FIG. 7, and FIG. 8B shows an F-F' cross section of the semiconductor device shown in FIG. 7. In FIG. 7, description of portions in common with those in FIG. 1B is omitted. In FIGS. 8A and 8B, description of portions in common with those in FIG. 2A is omitted.

[0149] The semiconductor device 2 according to the second embodiment shown in FIG. 7 includes third semiconductor regions 23 disposed between a plurality of gate regions 30 provided in the X direction, the third semiconductor regions 23 being offset with respect to each other in the X direction and the Y direction. That is, the plurality of third semiconductor regions 23 are provided in the Y direction, and at least one set of third semiconductor regions 23 among the plurality of third semiconductor regions 23 are disposed to be offset with respect to each other in the X direction, which means that at least one of the third semiconductor regions 23 is located in the positive direction of the X direction between the gate regions 30, and at least one of the third semiconductor regions 23 is located in the negative direction of the X direction between the gate regions 30. In other words, the third semiconductor regions 23 are not aligned in the Y direction.

[0150] Here, the expression “the third semiconductor region 23 is located in the positive (negative) direction of the X direction between the gate regions 30” means that the center of each of the plurality of third semiconductor regions 23 is located in the positive (negative) direction of the X direction with respect to a median line MM' connecting points at equal distances from two gate regions 30 adjacent to each other in the X direction. Among the third semiconductor regions 23 located in the positive direction of the X direction, an end in the negative direction of the X direction may be located in the negative direction of the X direction from the median line MM' as in the example shown in FIG. 7. According to this embodiment, it is possible to achieve a satisfactory contact by the plurality of plugs 40 provided in the Y direction even when the end of the third semiconductor region 23 in the negative direction of the X direction is located in the positive direction of the X direction from the median line MM'.

[0151] The gate region 30 includes a first gate insulating film 31a, a first gate electrode 32a, a second gate insulating film 31b, and a second gate electrode 32b. The first gate electrode 32a is covered with the first gate insulating film 31a. The second gate electrode 32b is covered with the second gate insulating film 31b. The first gate insulating film 31a and the second gate insulating film 31b are formed to be spaced apart from each other in the X direction.

[0152] The third semiconductor region 23 includes a region close to the first gate insulating film 31a and a region close to the second gate insulating film 31b.

[0153] That is, when comparing a distance in the X direction between the third semiconductor region 23 and the first gate insulating film 31a with a distance in the X direction between the third semiconductor region 23 and the second gate electrode 32b, there are a region where the former is shorter than the latter and a region where the former is longer than the latter among the third semiconductor regions 23.

[0154] In this manner, a plurality of plugs 40 are provided in the Y direction to be electrically connected to the third semiconductor regions 23 that are disposed to be displaced from each other in the X direction. Positions of the plurality of third semiconductor regions are offset in the second direction (in the X direction). Positions of the plurality of plugs are also offset in the second direction (in the X direction).

[0155] A length L1 of the plug 40 in the X direction is different from a length L2 in the Y direction. For example, $L1 < L2$, and $L1 \leq 2T_{\max}$. In order to improve flatness and fill the plugs 40 more satisfactorily by preventing generation of voids, it is preferable that $L1 \leq (5/3) T_{\max}$ or less. More preferably, $L1 \leq (4/3) T_{\max}$ or less.

[0156] FIG. 8A is a cross-sectional view taken along a line E-E' shown in FIG. 7. The third semiconductor region 23 is located near one of the gate regions 30 in the X direction. In other words, a distance between the third semiconductor region 23 and the first gate insulating film 31a is shorter than a distance between the third semiconductor region 23 and the second gate insulating film 31b.

[0157] The plug **40** is provided to penetrate an interlayer insulating film **50**, and electrically connects a first electrode **11** and the semiconductor substrate **20**. The plug **40** may be covered with a barrier metal **41**. The barrier metal **41** has a layered structure containing, for example, Ti and TiN. [0158] FIG. **8A** shows an example in which the plug **40** is formed on the third semiconductor region **23** near the center of the third semiconductor region **23**. On the other hand, the plug **40** may penetrate the third semiconductor region **23** in the Z direction and reach a second semiconductor region **22**. In addition, the plug **40** is not limited to being formed near the center of the third semiconductor region **23** in the X direction, but may be formed at a position closer to or farther from the first gate insulating film **31a**.

[0159] FIG. **8B** is a cross-sectional view taken along a line F-F' shown in FIG. **7**. The third semiconductor region **23** is located in the X direction near the gate region **30** on a side in the X direction opposite to that in the case of FIG. **8A**. A distance between the third semiconductor region **23** and the first gate insulating film **31a** is longer than a distance between the third semiconductor region **23** and the second gate insulating film **31b**.

[0160] The semiconductor substrate **20** and the first electrode **11** are electrically connected to each other via the plug **40**.

[0161] In the semiconductor device **2** of this embodiment, the plug **40** is disposed in accordance with the arrangement of the third semiconductor region **23**, and thus electrical connection between the third semiconductor region **23** and the plug **40** and electrical connection between the semiconductor substrate **20** and the first electrode **11** are improved.

[0162] First, description will be given by making a comparison with a case where electrical connection with the third semiconductor region **23** is attempted using a single plug **140** extending in the Y direction, as in the semiconductor device **102** of the second reference example shown in FIG. **6**. With the single plug **140** shown in FIG. **6**, when the third semiconductor region **23** is offset in the positive or negative direction of the X direction, there is a concern that a contact area between the plug **140** and the third semiconductor region **23** may be reduced. Furthermore, there is a concern that the electrical connection between the plug **140** and the third semiconductor region **23** may not be made at least partially.

[0163] Furthermore, as shown in FIG. **7**, when the third semiconductor region **23** includes portions that are offset in the positive and negative directions of the X direction, there is a concern that a contact with the third semiconductor region **23** may not be sufficiently made using a single plug **140** as shown in FIG. **6**. That is, when the plug **140** is formed to make a contact with the third semiconductor region **23** that is offset in one direction, it is difficult to make a contact with the third semiconductor region **23** that is offset in the other direction.

[0164] On the other hand, with the semiconductor device **2** of this embodiment shown in FIG. **7**, the arrangement of each of the plurality of plugs **40** can be adjusted in the X direction, making it possible to form the plugs **40** above each of the third semiconductor regions **23** that are offset in the X direction. Thus, it is possible to secure a reliable contact between the plug **40** and the third semiconductor region **23**.

[0165] For example, in FIG. **8A**, a case where a first semiconductor region **21** is an n-type drift layer, the second semiconductor region **22** is a p.sup.-type base layer, and the third semiconductor region **23** is an n.sup.+type emitter layer is considered. By applying a positive voltage to the gate electrode **32** with respect to the first electrode **11**, an n-type inversion layer is generated in the second semiconductor region **22**, and the semiconductor device **2** is set to be in an ON state. FIG. **8A** shows a structure in which the third semiconductor region **23** is located near the first gate electrode **32a** among the plurality of gate electrodes **32**. Thus, a current flowing in an ON state flows from the first semiconductor region **21** through the inversion layer generated mainly around the first gate insulating film **31a** to the third semiconductor region **23**.

[0166] The current flowing to the third semiconductor region **23** flows to the first electrode **11**, which is, for example, an emitter electrode, via the plug **40** electrically connected to the third

semiconductor region **23**. In this embodiment, the plug **40** is disposed close to one gate region **30** (the first gate insulating film **31a** and the first gate electrode **32a**) in FIG. **8A** in accordance with the arrangement of the third semiconductor region **23**. It is possible to secure a more reliable electrical connection between the third semiconductor region **23** and the plug **40** than when the plug **40** is disposed close to the gate region **30** (the second gate insulating film **31b** and the second gate electrode **32b**) where the third semiconductor region **23** is not closely located in FIG. **8A**. [0167] The same is true of the arrangement of the third semiconductor region **23** and the plug **40** shown in FIG. **8B**.

[0168] In this manner, the plug **40** is provided in accordance with the arrangement of the third semiconductor region **23**, and thus with the semiconductor device **2** of this embodiment, it is possible to improve electrical connection between the semiconductor substrate **20** and the first electrode **11** via the plug **40**. That is, with the semiconductor device **2** of this embodiment, the degree of freedom in designing the semiconductor device can be improved by diversifying a possible structure of the third semiconductor region **23**. For a predetermined layout of the third semiconductor region **23**, it is possible to expand the range of selection according to optimization of current control, including the arrangement of the plug **40** offset in the X direction between the gate regions **30**. It is possible to further improve the electrical characteristics by appropriately selecting the layouts of the third semiconductor region **23** and the plug **40**.

Third Embodiment

[0169] FIGS. **9** and **10** are an enlarged plan view and a cross-sectional view showing a semiconductor device **3** according to a third embodiment. FIG. **10** is a cross-sectional view showing a G-G' cross section of the semiconductor device shown in FIG. **9**. Description of portions in common with those in the first embodiment will be omitted as appropriate, and differences will be described.

[0170] In the semiconductor device **3** according to this embodiment, a gate electrode **32** provided in a gate region **30** is electrically connected to a plug **40**. In FIG. **9**, a plurality of plugs **40** are provided in the Y direction for the gate region **30** extending in the Y direction. In FIG. **9**, a gate wiring **13** and an interlayer insulating film **50** shown in FIG. **10** are omitted.

[0171] As shown in FIG. **9**, the plug **40** has a length L1 in the X direction and a length L2 in the Y direction. L1 is smaller than a length L7 of the gate electrode **32** in the X direction. In addition, for example, $L2 < L7$, and the gate region **30** is wider in the X direction. An interval in the Y direction between the plurality of plugs **40** formed in the Y direction is L3. For example, $L2 < L1$.

[0172] The plug **40** is electrically insulated from a semiconductor substrate **20** by a gate insulating film **31**.

[0173] FIG. **10** is a cross-sectional view taken along a line G-G' in FIG. **9**. As shown in FIG. **10**, the gate wiring **13** is electrically connected to the gate electrode **32** via the plug **40**. The plug **40** is made of a metal containing, for example, W (tungsten). The gate electrode **32** contains, for example, polysilicon. The gate electrode **32** is, for example, a gate electrode of an IGBT, and the gate wiring **13** is, for example, a gate wiring electrically connected to the gate electrode **32**. The potential of the gate electrode **32** can be controlled by applying a voltage to the gate wiring **13**.

[0174] The plug **40** may be covered with a barrier metal **41**. The barrier metal **41** has a layered structure containing, for example, Ti and TiN.

[0175] The semiconductor device **3** according to this embodiment may include a first electrode **11** provided on the interlayer insulating film **50** in a region separated from the region shown in FIGS. **9** and **10**. The first electrode **11** is, for example, an emitter electrode of an IGBT. The first electrode **11** is electrically connected to a third semiconductor region **23** of the semiconductor substrate **20** via a conductive member (not shown) provided in the interlayer insulating film **50**. The third semiconductor region **23** is, for example, an emitter region of an IGBT.

[0176] Furthermore, the conductive member connecting the first electrode **11** and the semiconductor substrate **20** may be a plug **40** different from the plugs **40** shown in FIGS. **9** and **10**.

In other words, a plug **40** different from the plugs **40** shown in the drawings may electrically connect the semiconductor substrate **20** and the first electrode **11** not shown in FIG. **9**. The semiconductor device **3** according to this embodiment may have a structure as shown in FIGS. **1B** and **2A** in a region not shown in FIG. **9**.

[0177] That is, the potentials of the semiconductor substrate **20** and the gate electrode **32** may be controlled via a plurality of plugs **40**. That is, by a structure not shown in FIGS. **9** and **10**, it is possible to control the potential of the semiconductor substrate **20** via the plugs **40** and separately control the potentials of the emitter electrode of the IGBT and the gate electrode via different plugs **40**.

[0178] With the semiconductor device **3** of this embodiment, the plurality of plugs **40** are spaced apart from each other in the Y direction, and thus it is possible to improve electrical connection between the gate wiring **13** and the gate electrode **32** regardless of the value of L7. For various shapes of the gate electrode **32**, it is possible to improve electrical connection by the plugs **40** and reliably form an inversion layer in the semiconductor substrate **20**.

[0179] Furthermore, a shorter length out of L1 and L2 is set to, for example, 2Tmax or less, and thus it is possible to prevent defective formation of the plugs **40** while preventing wafer warpage. It is possible to improve the reliability of the semiconductor device by preventing defective formation.

[0180] With the semiconductor device **3** of this embodiment, it is possible to form a plug **40** having a length L1 that is proportional to L7. As L7 increases, it is possible to increase a contact area between the gate electrode **32** and the plug **40**.

[0181] In addition, L7 may be within the range of an inequality $n \times 2T_{\max} + (n-1) \times L_{\min} < L7 < (2n+1) \times L_{\min}$ for a certain natural number n.

[0182] By increasing the range of selection of the value of L7, it is possible to increase the range of selection according to optimization of current control, for example, when the magnitude and distribution of a current flowing through the semiconductor device are controlled in accordance with the arrangement of the gate region **30**. It is possible to improve the electrical characteristics by appropriately selecting the value of L7.

[0183] Furthermore, it is possible to control the potential of the gate electrode **32** by electrically connecting the gate electrode **32** and the gate wiring **13** via the plug **40**. The potentials of the plurality of gate electrodes **32** arranged in the X direction can be individually controlled by different plugs **40**, and a multi-gate structure can be achieved by the plurality of plugs **40**. By individually controlling the potentials of the gate electrodes **32**, it is possible to reduce a loss during switching and further improve the performance of the semiconductor device.

[0184] In at least one embodiment described above, the plurality of plugs **40** provided in the Y direction each have a length L1 in the X direction and a length L2 in the Y direction, and at least one of L1 and L2 is formed short to be able to reduce stress on a wafer. Thus, a contact area between the plugs **40** and the semiconductor substrate **20** is made large by forming the other of L1 and L2 long, and thus it is possible to improve the performance of the semiconductor device for a wider range of the value of LG or L7. In addition, it is possible to prevent defective formation of the plugs **40** and improve the reliability of the semiconductor device.

[0185] In addition, it is possible to improve the degree of freedom in design for the range of the value of LG or L7 and improve the degree of freedom in design for the layouts of the third semiconductor region **23** and the plug **40**. It is possible to increase the range of selection of the value of LG or L7 or the layouts of the third semiconductor region **23** and the plug **40**. By appropriately selecting the value of LG or L7 or the layouts of the third semiconductor region **23** and the plug **40**, the electrical characteristics of the semiconductor device can be improved by optimizing current control.

[0186] Hereinafter, a manufacturing method for a semiconductor device will be described.

[0187] A manufacturing method for the semiconductor device **1** according to the first embodiment

will be described with reference to FIGS. **11** to **16**.

[0188] In the following description, FIGS. **11** to **16** are cross-sectional views corresponding to the A-A' cross section of the semiconductor device shown in FIG. **1B**. That is, FIGS. **11** to **16** show cross-sectional views at positions corresponding to the A-A' cross section of the semiconductor device shown in FIG. **1B**, which is a completed diagram, in the respective manufacturing processes.

[0189] In description of the manufacturing method, the formation of the plug **40** will be described in detail, while description of the formation of the gate region **30**, the semiconductor substrate **20**, and the second electrode **12** will be omitted since it is not significantly different from a general manufacturing method.

[0190] FIG. **11** is a cross-sectional view showing a process of providing the interlayer insulating film **50**. The semiconductor device **1** includes the semiconductor substrate **20** and the gate region **30** for which the manufacturing method is not described. The semiconductor substrate **20** contains, for example, Si, and an n-type semiconductor layer is formed, for example, by doping with N or P, and the p-type semiconductor layer is formed, for example, by doping with B. The gate region **30** includes the gate insulating film **31** and the gate electrode **32**. The gate insulating film **31** contains, for example, silicon oxide. The gate electrode **32** is, for example, polysilicon.

[0191] As shown in FIG. **11**, the interlayer insulating film **50** is provided on the semiconductor substrate **20** and the gate region **30**. The interlayer insulating film **50** is, for example, an oxide film containing silicon oxide which is formed by chemical vapor deposition (CVD).

[0192] Next, as shown in FIG. **12**, a resist **60** is provided on the interlayer insulating film **50**. The resist **60** is selectively formed, for example, by photolithography.

[0193] Next, as shown in FIG. **13**, an opening **70** is provided by excavating a portion where the resist **60** is not provided. A plurality of openings **70** are formed at intervals in the Y direction. The openings **70** are formed by, for example, reactive ion etching (RIE) or chemical dry etching (CDE). Some of the plurality of openings **70** penetrate the third semiconductor region **23** as shown in FIG. **13** and reach the second semiconductor region **22**. Some of them may also reach the second semiconductor region **22** without being in contact with the third semiconductor region **23** in a cross section different from the A-A' cross section shown in FIG. **13**.

[0194] Next, the resist **60** is peeled off, and a conductive film **80** is formed as shown in FIG. **14**. However, the barrier metal **41** may be formed before the conductive film **80** is formed. The barrier metal **41** has a layered structure containing, for example, Ti and TiN. The barrier metal **41** is formed by, for example, CVD. The conductive film **80** contains, for example, W (tungsten). The conductive film **80** is formed by, for example, CVD.

[0195] Next, as shown in FIG. **15**, the conductive film **80** is partially removed by etching back, and the remaining part is referred to as a plug **40**.

[0196] Finally, as shown in FIG. **16**, the first electrode **11** is provided on the plug **40** and the interlayer insulating film **50**. The first electrode **11** contains, for example, AlCu or AlSi. The first electrode **11** is formed by, for example, sputtering. In this manner, the configuration shown in FIG. **2A** is obtained.

[0197] In the manufacturing method described above, in the process of embedding the conductive film **80** in the opening **70** to form the plug **40**, for example, as described for the semiconductor device **1** according to the first embodiment, a plurality of plugs **40** are provided in the Y direction, and thus a contact area between the plug **40** and the semiconductor substrate **20** can be increased over a wider range of the value of an interval (LG shown in FIG. **1B**) between the gate regions **30**. Further, in the process of embedding the conductive film **80** in the opening **70** to form the plug **40**, the opening **70** has, for example, a rectangular shape in the XY plane, and thus it is possible to prevent generation of voids to achieve satisfactory embedding while reducing stress applied to the semiconductor substrate **20** by the plug **40**.

[0198] The embodiments are described above with reference to the specific examples. However,

the embodiments are not limited to the specific examples. That is, the specific examples in which the designs are changed by a person skilled in the art are also included in the embodiments as long as they have the characteristics of the embodiments. The elements of the above-described specific examples, their arrangements, materials, conditions, shapes, sizes and the like are not limited to illustrated ones and may be appropriately changed.

[0199] In addition, the elements of the above-described embodiments can be combined if technically possible, and the combined elements are included in the scope of the embodiments as long as they have the characteristics of the embodiments. It is understood that variations and modifications can be made by a person skilled in the art within the scope of the idea of the embodiments, and the variations and modifications are also included in the scope of the embodiments.

[0200] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure.

Claims

1. A semiconductor device comprising: a semiconductor substrate; an interlayer insulating film that is provided on the semiconductor substrate; a first electrode that is provided on the interlayer insulating film; a second electrode that is provided below the semiconductor substrate; a plurality of gate regions extending from the interlayer insulating film in a first direction, which is a thickness direction of the semiconductor substrate, to reach the semiconductor substrate, provided in a second direction intersecting the first direction, and extending in a third direction intersecting the first direction and the second direction; and a plurality of plugs located between the gate regions in the second direction, longer in the second direction than in the third direction, spaced apart from each other in the third direction, and electrically connecting the first electrode to the semiconductor substrate.
2. The semiconductor device according to claim 1, wherein a shape of each of the plugs in a plane including the second direction and the third direction is a rectangular or elliptical shape.
3. The semiconductor device according to claim 2, wherein the semiconductor substrate includes a first semiconductor region of a first conductivity type, a second semiconductor region of a second conductivity type which is provided on the first semiconductor region, a third semiconductor region of the first conductivity type which is selectively provided on the second semiconductor region, and a fourth semiconductor region of the second conductivity type which is provided between the second electrode and the first semiconductor region, and the gate region includes a gate electrode that is embedded in the gate region, and a gate insulating film that covers the gate electrode.
4. The semiconductor device according to claim 3, wherein the third semiconductor region includes a plurality of sub-regions, which are provided between a plurality of the gate regions and spaced apart from each other in the third direction.
5. The semiconductor device according to claim 4, wherein at least one of the plugs is in contact with the second semiconductor region and is provided between two of the sub-regions of the third semiconductor region.
6. The semiconductor device according to claim 5, wherein at least some of the plurality of plugs that are provided adjacent to each other in the third direction, are in contact with the third semiconductor region.

7. The semiconductor device according to claim 5, wherein at least some of the plurality of plugs that are provided adjacent to each other in the third direction, are each located between two of the sub-regions of the third semiconductor region.
8. The semiconductor device according to claim 1, wherein a depth D of each of the plugs in the first direction and a length L1 of each of the plugs in the second direction satisfy the relationship: $0 < D/L1 \leq 2$.
9. The semiconductor device according to claim 1, wherein each of the plugs contains W.
10. The semiconductor device according to claim 9, wherein the semiconductor device further comprises a barrier metal that covers each of the plugs and contains Ti and TiN.
11. A semiconductor device comprising: a semiconductor substrate; an interlayer insulating film that is provided on the semiconductor substrate; a first electrode that is provided on the interlayer insulating film; a second electrode that is provided below the semiconductor substrate; a plurality of gate regions extending from the interlayer insulating film in a first direction, which is a thickness direction of the semiconductor substrate, to reach the semiconductor substrate, provided in a second direction intersecting the first direction, and extending in a third direction intersecting the first direction and the second direction; and a plurality of plugs located between the gate regions in the second direction, spaced apart from each other in the third direction, and electrically connecting the first electrode to the semiconductor substrate; and a plurality of third semiconductor regions in the semiconductor substrate, provided between a plurality of the gate regions, and not aligned in the third direction.
12. The semiconductor device according to claim 11, wherein the third semiconductor regions are spaced apart from each other in the third direction.
13. The semiconductor device according to claim 11, wherein positions of the plurality of third semiconductor regions are offset in the second direction.
14. The semiconductor device according to claim 13, wherein positions of the plurality of plugs are offset in the second direction.
15. The semiconductor device according to claim 11, wherein some of the third semiconductor region are located on a first side of a line extending in the third direction, and some of the third semiconductor regions are located on a second side of the line that is opposite to the first side.
16. The semiconductor device according to claim 11, wherein one of the plurality of the gate regions includes a first gate insulating film and a second gate insulating film spaced apart from the first gate insulating film in the second direction, and some of the third semiconductor regions are closer to the first gate insulating film than the second gate insulating film, and some of the plurality of plugs are closer to the first gate insulating film than the second gate insulating film.
17. The semiconductor device according to claim 11, wherein each of the plugs contains W.
18. A semiconductor device comprising: a semiconductor substrate; an interlayer insulating film that is provided on the semiconductor substrate; a first electrode that is provided on the interlayer insulating film and is connected to the semiconductor substrate; a gate wiring that is provided on the interlayer insulating film to be spaced apart from the first electrode; a second electrode that is provided below the semiconductor substrate; a plurality of gate regions extending from the interlayer insulating film in a first direction, which is a thickness direction of the semiconductor substrate, to reach the semiconductor substrate, provided in a second direction intersecting the first direction, and extending in a third direction intersecting the first direction and the second direction, each of the gate regions including a gate electrode that is embedded in the gate region and a gate insulating film that covers the gate electrode; and a plurality of plugs that are spaced apart from each other in the third direction and electrically connect the gate wiring to the gate electrode of each of the gate regions.
19. The semiconductor device according to claim 18, wherein the plugs are longer in the second direction than in the third direction.
20. The semiconductor device according to claim 18, wherein each of the plugs contains W.

